Chankyo Peter Kim

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Ann Arbor, MI · Los Altos, CA

Research focus: Robot learning · vision-language-action (VLA) · policy post-training · sim-to-real

Education

University of Michigan, Ann Arbor 2024 – Present
Ph.D. in Robotics; Advisor: Prof. Maani Ghaffari *Expected May 2028*

Seoul National University 2022
B.S. in Mechanical and Aerospace Engineering; Summa Cum Laude *GPA: 4.06/4.3; top 3 in department*

Research Experience

Toyota Research Institute (TRI) May 2026 – April 2027
Research Scientist Intern, World Foundation Model Team *Los Altos, CA*

- **Diffusion-policy reinforcement learning (RL)** (patent filed). Fine-tuned a diffusion driving policy, raising **full-route completion from 17% to 80%** on unseen routes. Built the training and evaluation framework adopted by TRI's end-to-end driving team.
- **ActionGuide: multimodal LLM post-training** (co-author; under review at ICLR 2027). Fine-tunes multimodal LLMs to check candidate driving paths and select a fallback (e.g., stop or turn) when all proposals are rejected. Evaluated in AlpaSim without retraining the planner.
- **VLA evaluation in generated scenes** (AnyWorld × PolaRiS). Developing an Isaac Sim pipeline to test VLA policies in generated 3D scenes; real-robot validation on TRI's DROID station is planned.
- **AnyWorld: vision-language foundation model for robot learning** (lead author). Jointly trained on video, depth, and language using **3,000+ video hours** plus indoor robot scans, on up to **32 GPUs**. Developing 3D Gaussian dynamics to predict scene changes after robot actions for closed-loop planning.

University of Michigan Sep. 2023 – Present
Graduate Student Researcher; Advisor: Prof. Maani Ghaffari *Ann Arbor, MI*

- **RoboBelief: VLA post-training and benchmark** (joint with Prof. Xingxing Zuo, MBZUAI). Fine-tuned X-VLA 0.9B with LoRA and a GRU filter that estimates hidden object locations; built a MuJoCo/LIBERO benchmark. In closed-loop simulation after 16 or 32 human actions, selected the most likely bowl location on its first choice in **160/160 trials vs. 75%** for a RoboMME-style memory VLA. Changing only the belief input switched the robot's choice in 40/40 tests.
- **Gaussian World Model and planning.** Modeled rigid-body motion with 3D Gaussians and integrated a sampling-based planner for closed-loop object pushing. Reached **69% success per attempt** across 24 *Pushing Gaussians* tasks (within 30 mm of the target).
- **GINIO: inertial odometry (CoRL 2026; patent filed).** Developed an $\mathrm{SO}(3)$-equivariant neural odometry interface. After remounting a Fetch robot's IMU in unseen orientations, reduced absolute trajectory error from **8.15 to 0.50 m** without retraining.
- **E3DGS: equivariant 3D Gaussian representations.** Built an $\mathrm{SE}(3)$-equivariant model of Gaussian geometry and view-dependent appearance with masked pretraining. Reached **95.2%** ModelNet10 classification accuracy under unseen rotations and translations, vs. **18.3%** for Gaussian-MAE.
- **Reductive Lie Neurons (ICML 2026).** Developed $\mathrm{GL}(n)$-equivariant networks for matrix and Lie-algebra features, including covariance and Gaussian shape matrices. PyTorch code.

Bear Robotics Jan. 2022 – Apr. 2022
Research Intern, Robotic Automation Team *Redwood City, CA*

- **3D SLAM for robotic mapping.** Built a 3D SLAM system with factor-graph optimization and camera-motion constraints; **adopted in Bear Robotics' mapping product**.

Technical Skills

Policy post-training: closed-loop reinforcement learning (RL), diffusion policies, VLA fine-tuning (X-VLA, LoRA), recurrent belief filtering (GRU), vision-language-action (VLA) policies (π_0 family), MPPI/CEM planning.
Simulation & evaluation: MuJoCo/robosuite/LIBERO, Isaac Sim, generated-scene policy evaluation, closed-loop policy evaluation, world-model benchmarking, MetaDrive.
World models & representations: vision-language foundation models, joint video–depth–language training, action-conditioned latent dynamics, 3D Gaussian Splatting (gsplat), equivariant networks on Lie groups.
Programming & systems: Python, C++, PyTorch, JAX, multi-GPU training, Hugging Face model packaging.

Selected Publications & Manuscripts

GINIO: A Geometric SO(3)-Equivariant Interface for Neural Inertial Odometry
Kim, C., Zhu, M., Lin, T.-Y., Rattan, A., Ghaffari, M.
CoRL 2026. [arXiv]

Equivariant Neural Networks for General Linear Symmetries on Lie Algebras (Reductive Lie Neurons)
Kim, C.*, Zhao, S.*, Zhu, M., Lin, T.-Y., Ghaffari, M.
ICML 2026. *Equal contribution. [Paper] [Code] [Project]

ActionGuide: Learning Transferable Action-Grounded Reasoning for Closed-Loop Driving
Wei, Y., Wu, J., **Kim, C.**, Liu, H., Joshi, R., Ghaffari, M., Patel, V. M., McAllister, R. T., Zeng, Y.
Under review at ICLR 2027. [Manuscript]

A World Model That Builds Any World to Scale
Kim, C., Guizilini, V. C., Ghaffari, M.
Under review at CVPR 2027. [Manuscript]

E3DGS: Unified Geometric–Photometric Equivariance for 3D Gaussian Splatting via Color-as-Geometry Embedding
Kim, C., Ghaffari, M.
Preprint. [arXiv]

Reductive Lie Neurons: A Framework for $\mathrm{GL}(n, \mathbb{R})$-Equivariant Deep Learning
Kim, C., Zhu, M., Ghaffari, M.
RSS Workshop on Equivariant Systems, 2025. [Workshop]

Patent filings: Closed-loop policy training and evaluation (TRI, 2026); equivariant inertial odometry (University of Michigan).

Selected Honors

TartanIMU Challenge Award, IROS 2026 Workshop "Beyond Exteroception: Interoceptive Perception for Resilient Robotics" — ENIO Team, 2026.

Kwang and Kook-Wha Koh Science and Engineering Scholarship, SNUAA-MI — 1 of 5 recipients, 2026.

Rackham Doctoral Intern Fellowship, University of Michigan, 2026.

SK hynix Global Researcher Scholarship (2026); **Qualcomm Innovation Fellowship Finalist** (2025); **Daishin Graduate AI Research Fellowship** (2024).

Sinyang Foundation Scholarship (2020–2022); SNU Academic Excellence Scholarship (four semesters, 2015–2016); Samsung Humantech Paper Award, Bronze (2014).

Academic Service, Teaching & Outreach

Reviewing: ICML, CoRL (2026); IEEE T-RO, ICRA, IROS (2024–2025).
Invited challenge presentation: ENIO, IROS 2026 Beyond Exteroception Workshop, 2026.
Talks: Invited seminar, "Why Equivariance? Foundation Models in Robotics," SNU (Dec. 2025); "Equivariance on Uncertainty and Reductive Lie Neuron Design," CVPR 2025.
Teaching: Graduate Student Instructor, University of Michigan (2024–2025): Computational Symmetry in AI & Robotics, Mobile Robotics, and Self-Driving Cars.
Mentorship: Three MS researchers working on action-conditioned world modeling, neural VIO/SLAM, and neural inertial odometry.
Educational materials: Four books on AI, Python, robotics, and engineering education (2021–2023).

Leadership

OUTTA Jan. 2021 – Present
Co-founder, AI/Robotics Education Non-profit *Seoul, KR*
Co-founded AI bootcamps serving **3,000+** underprivileged students and early-career developers. [Website]